LEFT ADJOINTS PRESERVE REPRESENTABLE COLIMITS AND RIGHT ADJOINTS REPRESENTABLE LIMITS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3705-3724, following the statement at 3688-3703.

Editorial note. The dual second part (line 3723, phrase: a similar argument works for the other statement) is retained verbatim as a mirror-level omission; the executed part (1) covers its dual.

Allowed standard prerequisites (editorial).

- Adjunction bijection. definition-adjoint, cited at statement line 3690. Supplies the first and third equalities of the chain at proof lines 3710-3719.
- Morphisms from a colimit as compatible systems. remark-limit-colim, cited at proof lines 3706-3708. Supplies the second equality Mor(colim M_i, v(Y)) = lim Mor(M_i, v(Y)) and the reading of the chain as the colimit universal property.

Source reference. Original definition 0037: sources/stacks-categories/pinned-original-source.tex, source lines 3561–3561. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 002H: sources/stacks-categories/pinned-original-source.tex, source lines 1559–1559. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let u be a left adjoint to v as in Definition 0037.*

(1) *Suppose that $M : \mathcal{I} \to \mathcal{C}$ is a diagram, and suppose that $\operatorname{colim}_{\mathcal{I}} M$ exists in $\mathcal{C}$. Then $u(\operatorname{colim}_{\mathcal{I}} M) = \operatorname{colim}_{\mathcal{I}} u \circ M$. In other words, u commutes with (representable) colimits.*

(2) *Suppose that $M : \mathcal{I} \to \mathcal{D}$ is a diagram, and suppose that $\lim_{\mathcal{I}} M$ exists in $\mathcal{D}$. Then $v(\lim_{\mathcal{I}} M) = \lim_{\mathcal{I}} v \circ M$. In other words v commutes with representable limits.*

Proof. A morphism from a colimit into an object is the same as a compatible system of morphisms from the constituents of the limit into the object, see Remark 002H. So

$$\begin{aligned} \operatorname{Mor}_{\mathcal{D}}(u(\operatorname{colim}_{i\in\mathcal{I}} M_i), Y) &= \operatorname{Mor}_{\mathcal{C}}(\operatorname{colim}_{i\in\mathcal{I}} M_i, v(Y)) \\ &= \lim_{i\in\mathcal{I}^{opp}} \operatorname{Mor}_{\mathcal{C}}(M_i, v(Y)) \\ &= \lim_{i\in\mathcal{I}^{opp}} \operatorname{Mor}_{\mathcal{D}}(u(M_i), Y) \end{aligned}$$

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proves that $u(\mathrm{colim}_{i \in \mathcal{I}} M_i)$ is the colimit we are looking for. A similar argument works for the other statement. $\square$